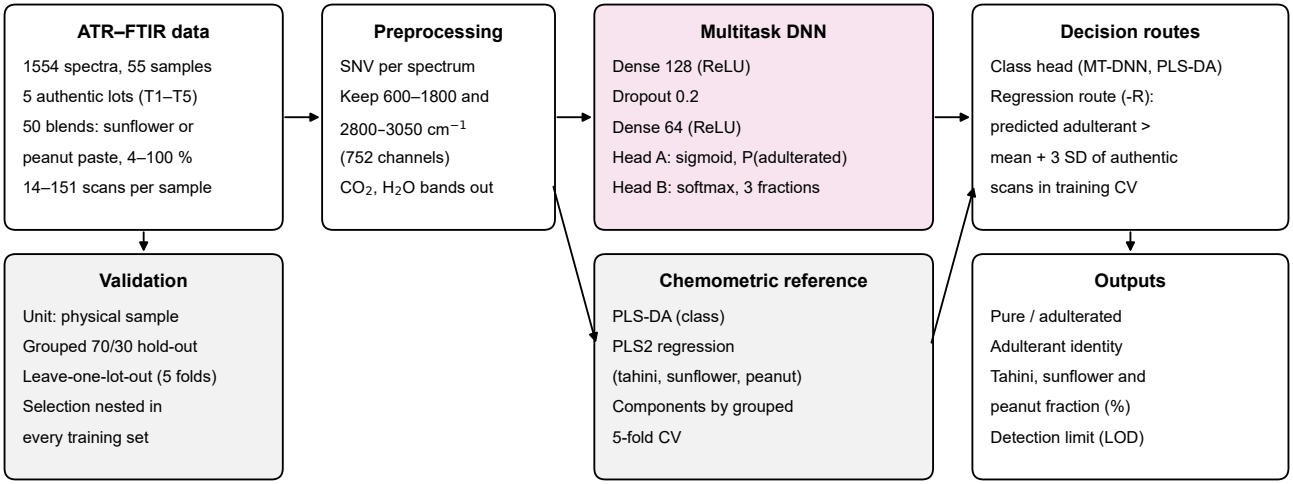

ATR–FTIR data
1554 spectra, 55 samples
5 authentic lots (T1–T5)
50 blends: sunflower or
peanut paste, 4–100 %
14–151 scans per sample
Preprocessing
SNV per spectrum
Keep 600–1800 and
2800–3050 cm−1
(752 channels)
CO2, H2O bands out
Multitask DNN
Dense 128 (ReLU)
Dropout 0.2
Dense 64 (ReLU)
Head A: sigmoid, P(adulterated)
Head B: softmax, 3 fractions
Decision routes
Class head (MT-DNN, PLS-DA)
Regression route (-R):
predicted adulterant >
mean + 3 SD of authentic
scans in training CV
Validation
Unit: physical sample
Grouped 70/30 hold-out
Leave-one-lot-out (5 folds)
Selection nested in
every training set
Chemometric reference
PLS-DA (class)
PLS2 regression
(tahini, sunflower, peanut)
Components by grouped
5-fold CV
Outputs
Pure / adulterated
Adulterant identity
Tahini, sunflower and
peanut fraction (%)
Detection limit (LOD)